Continual Learning for SAR Target Incremental Detection via Predicted Location Probability Representation and Proposal Selection

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***Abstract*— The gradual increase of synthetic aperture radar (SAR) imagery often accompanies the appearance of new targets, but traditional detection frameworks can only detect existing target classes and cannot detect new ones. Typically, we must update the model using both new and old data, which puts a strain on storage and computation, but if we only update with new data, the detection performance on old classes will suffer dramatically. For this reason, this article proposes to use the continual learning (CL) method to solve the problem of SAR target incremental detection. Mainstream CL methods generally consider localization to be a class-irrelevant function, however, this strategy is unsuitable for SAR imagery with significant background changes, leading to poor detection performance. Addressing the above issues, this article proposes a CL object detection (CLOD) method, with an overall framework based on knowledge distillation. The focus of the methodology consists of two parts: first, we introduce the predicted location probability representation (PLPR) method, by using spatial discretization and segmented probabilistic statistics, to transform the localization results into probability distributions, thus allowing the localization function to participate in the CL process; second, we design a proposal selection strategy according to the background characteristics of SAR images, which improves the quality of the proposals during the knowledge review to further optimize the learning effect. Experiments on the latest multiclass SAR target detection dataset MSAR-1.0, show that our method is able to learn new knowledge with less performance penalty for old classes than other methods. In multiple data incremental settings, our method provides a 2%–11% performance improvement over numerous common methods.**

***Index Terms*— Catastrophic forgetting, continual learning (CL), predicted location probability representation (PLPR), proposal selection strategy, synthetic aperture radar (SAR) target detection.**

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I. Introduction

SYNTHETIC aperture radar (SAR) target detection is an important application in the field of remote sensing, which is widely used in ocean monitoring. With the development of radar technology, the number of SAR images available in recent years has risen, thereby raising the requirements for the performance and practicality of SAR target detection algorithms. Traditional SAR target detection algorithms rely on hand-designed features and classifiers, which must be manually designed through image gray level, target scale, and other unique morphological features. The algorithm process can be divided into three stages: region selection, feature extraction, and classification [1]. The purpose of region selection is positioning and feature extraction is to extract the classification features of the region on the basis of region selection, which is the core of the target detection method. Commonly used features include scale-invariant feature transform (SIFT) [2], [3] and histogram of oriented gradient (HOG) [4].

However, these features do not perform well in complex scenarios. For example, factors such as the shape, size, and surface characteristics of the target make the scattering model more complex, making accurate localization and recognition difficult [5]. The attenuation and distortion of radar signals caused by atmospheric conditions can cause blurring and deformation of the SAR image, affecting target detection capability. The ground coverage and clutter interference make it challenging to accurately extract the target features [6], especially in cases where the backgrounds are similar or overlapping. Even the most frequently used and best CFAR detectors are also affected by background statistical regions. Therefore, with the development of a convolutional neural network, the target detection algorithm based on deep learning has gradually become a mainstream direction in the field of SAR target detection. For SAR target detection, due to the limitation of the dataset, previous works mostly focus on single-class target detection, such as SAR ship detection [7], [8], [9]. With the gradual enrichment of SAR images in recent years and the proposal of multiclass SAR target detection datasets, many researchers have turned attention to multiclass SAR target detection [10].

In realistic scenarios, sensors will continuously capture new targets and types, so the database will be constantly supplemented with new types of data. Learning new data

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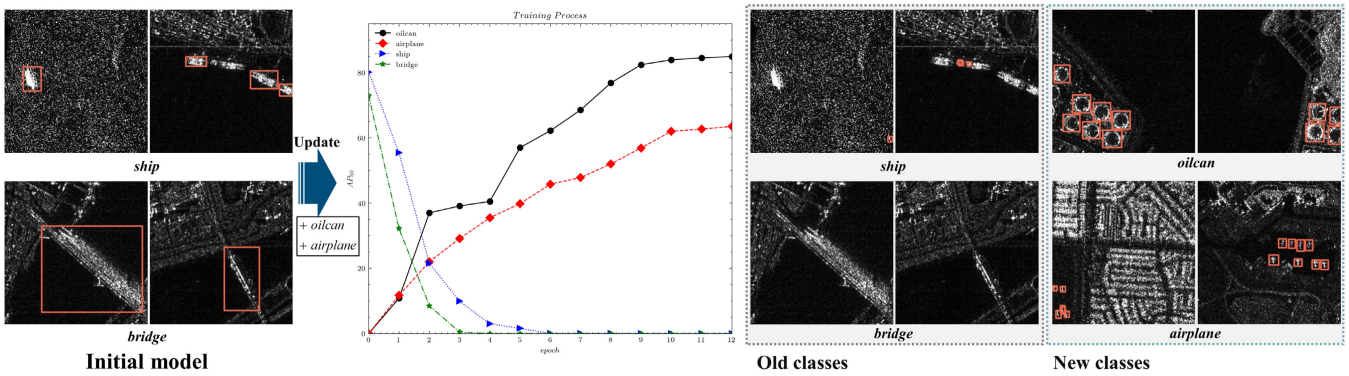


Fig. 1. Example of catastrophic forgetting in SAR target detection. The initial model is able to accomplish the task of detecting ships and bridges, but when we update the model with only new data (oilcan and airplane), its detection ability on the old classes decreases sharply.

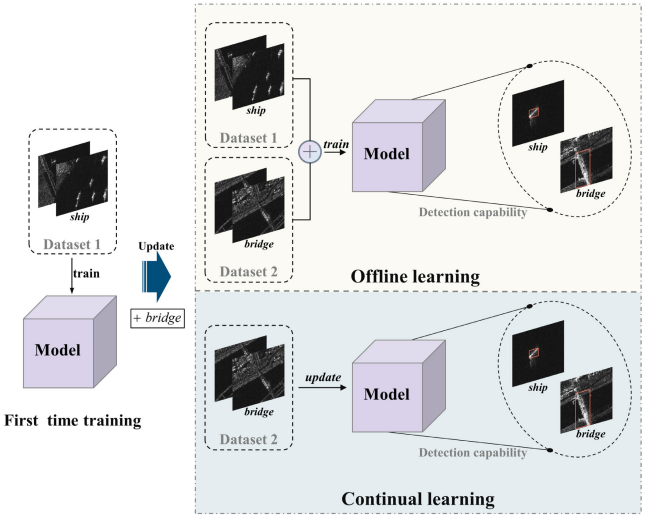


Fig. 2. Difference between offline learning and CL, offline learning requires repeated use of old images during the updating process, while CL only requires the use of new images.

requires at a minimum the following processes: manually identifying new classes, labeling new data, and updating the model. In the general offline training mode, all historical and new data are needed for model updating. If only using new data to update, the model's ability to detect old classes will be reduced (like the example shown in Fig. 1), which is called "catastrophic forgetting" [11], [12].

Obviously, the general offline training method requires more storage space and time, and continual learning (CL) is more suitable for this scenario of gradually increasing data. CL methods only need new data or a small amount of old data to update the existing model, which can effectively combat "catastrophic forgetting" while avoiding repeated old samples, the difference in this process is shown in Fig. 2. Due to the practicality of CL methods, many object detection methods based on deep learning have been extended from offline learning to CL.

CL was first applied to object classification to solve the problem of the constant addition of new classes. The representative works include the CL method based on replay mechanism incremental classifier and representation learning (iCaRL) [11] and its improved algorithm [13], [14], which achieved the purpose of combating catastrophic forgetting by retaining part of the representative old images. LwF [12] based on distillation [15] and its improved algorithm [16], [17] do not use new data at all when training new data, and use old knowledge to impose constraints on the model through knowledge distillation to protect the old knowledge from being completely covered by new knowledge. Fast-incremental learning for object detectors (ILOD) [18] pioneered the introduction of LwF into the field of object detection, and used knowledge distillation to realize class incremental on the basis of Fast region-based convolutional neural networks (RCNN) [19]. However, because Fast RCNN is a nonend-to-end object detection network and has poor practicability, some researchers soon introduced the CL method into the end-to-end detection network. Representative works include Faster-ILOD [20] based on the two-stage detector Faster RCNN [21], elastic response distillation (ERD) [22] based on one-stage detector generalized focal loss (GFLv1) [23] and selective and inter-related distillation (SID) [24] based on anchor-free detector CenterNet [25]. In addition to the methods based on knowledge distillation, other methods have also been used to achieve continual object detection in recent years. Representative works include the meta-learning-based method ONCE [26], feature-preserving methods rodeo [27], and so on. Overall, relatively less work has been conducted on CL in the field of object detection, and the vast majority of the work has been achieved through knowledge distillation.

As SAR images become more abundant, how to continuously update the model in the process of increasing data will become a challenge. Although the above methods have achieved advantages in the field of general target detection, but lack consideration of the characteristics of SAR imagery. Unlike target classification, detectors need to achieve both target localization and recognition, which makes incremental target detection methods additionally deal with localization tasks. For localization, previous work considered the localization of the target detector to be a "class-irrelevant" function [20], [24], and therefore, during CL, isolating or fine-tuning the parameters of the regressor, the detector was still able to localize the target. This practice results in a decrease in localization accuracy but within acceptable limits. However, this practice requires two prerequisites: first, the initial training set is sufficiently rich in background information; second, the background environment should not be too different between the old and new classes of images. Obviously, the available SAR target detection data makes

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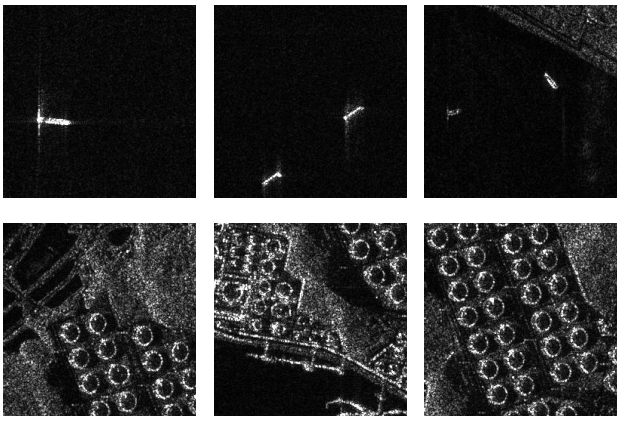


Fig. 3. Examples of SAR images. The first row shows SAR ship images and the second row shows SAR oilcan images.

it difficult to fulfill the above conditions. The targets and environments in SAR images are extremely correlated and usually have a single background for similar targets. For example, as shown in Fig. 3, the background of the SAR ship image is a dark-colored sea, while the background of the oilcan is a land covered with bright spots, the difference in background information is too large. This has caused prior methods to be poorly effective in the field of SAR target incremental detection.

To solve the above problem, this article proposes a CL method for SAR target detection, with the overall framework based on knowledge distillation. Also inspired by localization distillation [28], the predicted location probability representation (PLPR) method is proposed to transform the regression output, which in turn engages the regression output in the process of knowledge distillation and enhances the robustness of the CL method in facing SAR images.

At the same time, we found that the choice of proposals will significantly affect the effectiveness of CL in the process of knowledge distillation [22], [29]. Due to the simplicity of the characteristics of the SAR target itself, proposals that behave similarly between the reference and existing models are more likely to confuse the results. Conversely, proposals with differential performance contribute more to CL. Therefore, this article proposes a proposal quality evaluation criterion, based on this criterion together with the quality balancing ideology [30], the proposal selection strategy is designed. This strategy is able to screen out proposals with higher contributions for better performance during the SAR target detector's CL process.

In the subsequent section of this article, we first summarize the previous related work, then detail each part of the method in Section III, finally verify the effectiveness of our method on the latest multiclass SAR target detection dataset MSAR-1.0 [10]. The contributions throughout the article are as follows.

1) We analyze the shortcomings of prior CL methods for SAR target detection, and show that probabilizing the detector scored-proposals has a positive impact on the learning effect.
2) For the problem of decreased localization ability caused by the increase in SAR images with large scene differences, we developed the PLPR by applying the CL framework. The method can convert the regression output into a probability distribution and thus incorporate the localization information into the CL process, thereby mitigating the decrease in localization ability during CL.
3) For the characteristics of SAR image targets and backgrounds, the proposal selection strategy was proposed. The method assists in the knowledge review process by evaluating the quality of proposals in SAR imagery, combining the quality-balance concept with an evaluation guideline. By reducing the number of useless proposals introduced into the CL process, the CL method becomes more applicable to SAR images.

II. Related Work

In this section, we first introduce representative works in the field of CL, and then further summarize the CL methods in the field of object detection.

A. Continual Learning

How to deal with catastrophic forgetting is the main challenge of CL. Currently, there are numerous works that have investigated this issue, but most of them focus on relatively straightforward classification problems since there is currently little research that uses it for object detection. The widely used CL method can be separated into two paradigms: the replay method and the regularization method, depending on how old knowledge is stored and used.

Replay-based methods [11], [12], [13], [14], [31], [32] prevent the forgetting of old knowledge by keeping representative images of old tasks and using them to train with new images. iCaRL [11] is considered as the most classical replay-based method. It assumes that the samples closer to the feature mean-value-center are more representative, and retain a part of representative old data for each class. Other improved algorithms based on iCaRL include [13], [14], [31], [32] and these methods improve the iCaRL algorithm from different perspectives to alleviate catastrophic forgetting. In addition, there are also some works that introduce the idea of GAN into CL. For example, [33] uses the GAN network to generate old data, which effectively avoids the potential data privacy problem of the replay method. However, this method is equivalent to indirectly storing old data by storing model parameters, and it needs to consider the generation effect of the generation network itself, so this kind of method is not ideal. In summary, the main drawbacks of replay-based CL methods are the need for additional storage space for remembering old knowledge and the possibility of data privacy leakage.

The main idea of regularization-based methods [34], [35], [36], [37] is to protect old knowledge from being overwritten by new knowledge by imposing constraints on the new task. LwF [12] is one of the most representative CL methods, which draws on the idea of knowledge distillation and introduces distillation loss to make the output of the old layer of the new model on the new task similar to that of the old model, so as to achieve the purpose of review. Since then, many researchers have improved LwF. The representative methods include the low-dimensional feature mapping algorithm proposed by

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Rannen et al. [16] and the parameter constraint algorithm based on the Bayesian framework proposed by Tian et al. [29].

For SAR target recognition, several works have introduced CL methods, and most of them are based on replay. Representative works such as [38], [39], and [40], these methods focus on how to select representative samples from old images and differ in the selection strategy. References [38] and [39] applying CL to traditional target recognition methods, [38] finds the boundary of different classes on the feature space, and keeps the samples on the boundary as representative samples. Furthermore, the work [39] optimizes the boundary-sample selection criterion and proposes a boundary reconstruction method. Li et al. [40] calculated the anchored-class-center in the feature space, and then combined it with the clustering algorithm to select representative samples. In addition to the above replay-based methods, regularization-based methods have also been proposed in recent years [41], [42], [43], [44]. For example, the latest work [44] designed a multilevel adaptive knowledge distillation network to combat catastrophic forgetting in SAR continual target recognition. Overall, CL methods based on replay are still the mainstream methods in the field of SAR target recognition. This is due to the fact that images used for recognition are small in size, take up less storage, and do not require information about the target's position in the image, so keeping few samples can be used to satisfy the learning requirements while reducing the storage pressure. However, the images used for detection do not satisfy the above conditions.

B. CL Object Detection (CLOD)

For this article, our work focuses on a CL method that will be applied to SAR target detection. Compared with CL in classification, continual object detection is more challenging, and the corresponding research is much less. Fast-ILOD [20] applies LwF to the object detector Fast RCNN, which introduces CL into the field of object detection for the first time. The reason why the author uses the nonend-to-end network Fast RCNN is that its localization function does not rely on parameter sharing, so it can approximately transform the detection problem into a classification problem, but this reduces the practicability of the method. Since then, researchers have introduced CL to end-to-end networks. Faster-ILOD [20] applies LwF to the two-stage end-to-end network Faster RCNN by exploring the robustness of RPN. Peng et al. [24] proposed the SID method, through knowledge distillation and parameter isolation, successfully combines CL with anchor-free detector CenterNet. The latest work [22], borrowed from the work localization distillation [28] in the field of knowledge distillation, implements the CL task on the one-stage detector GFLv1 for the detector head response.

III. Method

Problem Definition: D_0 and D_1 denote the dataset containing the classes C_0 and C_1, respectively, and $D_0 \cap D_1 = \emptyset$. Given a detection model M_0 that has been well-trained on dataset D_0, our task is to update M_0 with only new data D_n to obtain a new model M_1 that performs well on both C_0 and C_1 images.

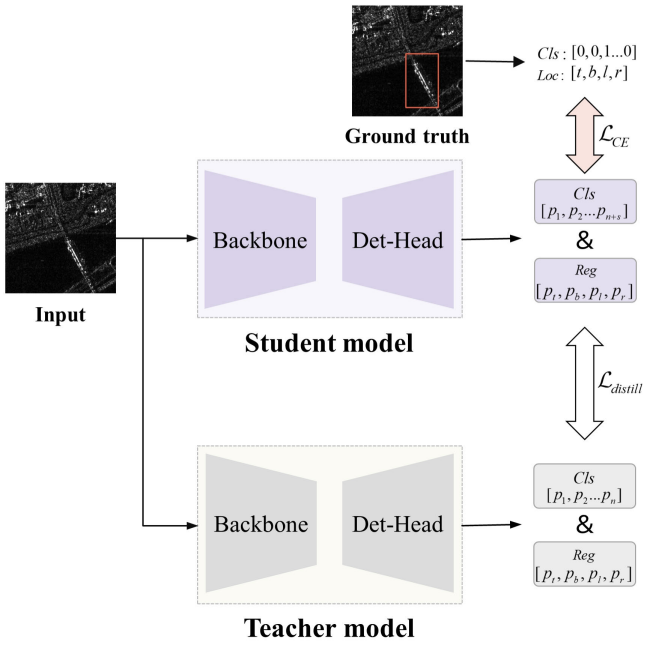


Fig. 4. Training process of CLOD. (Top and bottom) Learning process of the student model and the teacher model, respectively. The student model learns new classes through regular training while learning old knowledge from the teacher model through knowledge distillation.

In order to accomplish the above tasks, this work proposes an SAR target CL detector based on knowledge distillation. The training process for CL is shown in Fig. 4. Before model updating, we freeze the previous model, and the frozen model (also called the teacher model) is used as old knowledge in the updating process. The whole updating process can divided into two parts, the general training part and the knowledge review part. General training is the offline training process that gives the model the ability to detect new classes. Knowledge review is an anti-forgotten process because the back-propagation process will adjust the relevant parameters of the whole network, while the student model, by mimicking the output of the teacher model, is able to prevent the old knowledge from being completely overwritten by the new. The methods in the article all operate on the process of knowledge review, so this part is the focus of the article.

In the subsequent sections, we first introduce the overall framework in Section III-A, and then detail the proposal selection strategy and predicted location probability representation (PLPR) in Sections III-B and III-C, respectively.

A. Overall Framework

Unlike classification, the object detection network needs to find the location of the target in the image and judge its category, these two functions are done by the classification branch and the regression branch of the detector head. The purpose of this approach is to make the detector perform well in two different functions. Similarly, our CLOD algorithm must consider both classification and localization.

The overall framework is shown in Fig. 5. The new data will be fed into both the teacher model and the student model,

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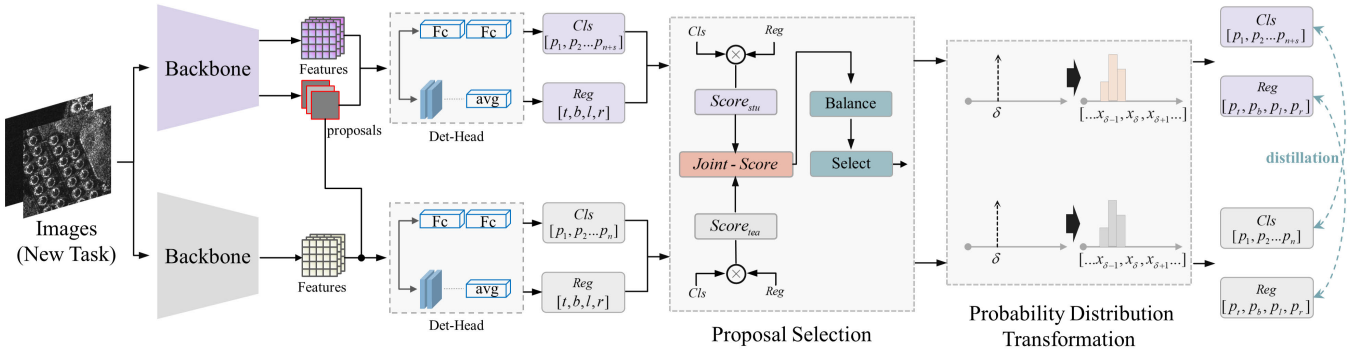


Fig. 5. Overall framework of our method. After obtaining the scored-proposals of the two models, the proposal-set for learning is first selected based on the different performances in the two models, and then the regression outputs of each proposal are transformed into probability distribution through the PLPR method and softness, further knowledge distillation is carried out.

and after the backbone, the network outputs feature maps and generates the initial proposals. Next, we chose to send the proposals from the student model to the teacher model at the same time, and then both models sent their own feature maps and the same proposals to their respective Det-head for the next step in the process. The reason for doing this is the need to ensure both models are processing the same objects. Furthermore, the Det-head will output scores which are the results of the judgment of the class and location of each proposal. The classification score is in the form of a set of probability values, of the length of the number of categories, representing the probability of each class; The localization score is in the form of a set of directed distance values, representing the position and size of the target.

Since the proposals are dominated by negative information, generally object detection and CLOD methods do not use all the proposals for learning. After obtaining the scores, we first measured the quality of the proposals by their different performances in the two models based on the SAR image characteristics and then selected the proposal-set that was most valuable for the knowledge review by combining them with the principle of quality balance.

After obtaining proposal scores, we use the PLPR method to convert the results into a probability distribution that can be used subsequently. Each score contains two parts, classification and regression. Since the classification itself is the probability distribution, the PLPR method focuses on the regression output. After obtaining the transformed scores, the network first softens them and then proceeds to knowledge distillation between the frozen model and the current model. Similarly, the knowledge distillation involves classification distillation and regression distillation, specific details will be elaborated in subsequent sections. In summary, the final loss can be defined as

$$\text{Loss} = \text{Loss}_{\text{model}} + \lambda_1 \text{Loss}_{\text{dis_cls}} + \lambda_2 \text{Loss}_{\text{dis_reg}} \tag{1}$$

where $\text{Loss}_{\text{model}}$ is the loss of the student detection network for new classes, including classification loss and localization loss under regular training; $\text{Loss}_{\text{dis_cls}}$ is the distillation loss of the classification branch; and $\text{Loss}_{\text{dis_reg}}$ is the localization distillation loss of the regression branch. The balancing weights, λ_1 and λ_2, are altered during training according to the training effect.

B. Proposal Selection

For the knowledge distillation-based CLOD method, the scores obtained from pure-background proposals received low confidence in both the teacher model and the student model, which contribute little to CL. Conversely, if a proposal shows high confidence in both the teacher model and the student model, then its likely to be a negative proposal with high similarity to the new class objectives, which can easily confound the judgment of the new class. In contrast, proposals that show differences between the two models contribute significantly to CL. For example, a proposal containing a noisy target with characteristics similar to the target of the old class, which is judged as background (nontarget) by the student model, may be predicted as the target of the old class by the teacher model, its score will show differences between the two models. For CL without old data, this is a proposal with high contribution, because in this case, it can act like a "pseudo-label."

The background information of SAR images is not as abundant compared to that of optical images, this results in large pure background areas and bright spot-like noise. As a consequence, the model for SAR target detection is very sensitive to the background information. Proposal-set with differentiation contributes much more to CL than simple negative samples, so using them can enhance the learning effect. Based on the above idea, this article defines the difference in scores between the teacher model and the student model from the same proposal as the joint-quality score (subsequently shortened to quality score), and the quality scores are calculated from the accuracy of the classification and localization results. Based on the above ideas, this article designs a proposal selection strategy in order to select the proposal-set with the highest quality during the knowledge review, and the detailed calculation process is as follows.

First, the scores obtained after a proposal has passed through the teacher and student networks are denoted as R_t and R_s, and each score contains predictions of category and position, denoted as p and b. Then, R_t and R_s can be denoted as $R_t = \{p_t, b_t\}$ and $R_s = \{p_s, b_s\}$.

Furthermore, the quality scores of the student model and teacher model are computed separately, incorporating classification and localization accuracy. The calculation is similar to object detection, which requires comparing the score with the ground truth. Specifically, for the teacher model, instead

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of the ground truth, outputs from the old class portion of the student model are used, that is because the teacher model only contains knowledge of the old class and the classification result is bound to be an incorrect result, thus using the student model output is much more smoothing. Therefore, the classification score of the student model needs to be processed prior to calculation by taking the old part values and normalizing, noted as $\text{soft}(p_t)$. Denote the quality score of the teacher network and the student network as x_t and x_s, respectively, and they are computed as

$$\begin{aligned} x_t &= \text{KL}(p_s, \text{soft}(p_t)) \cdot \text{IoU}(b_s, b_t) \\ x_s &= \text{MAE}(p_s, p_{\text{GT}}) \cdot \text{IoU}(b_s, b_{\text{GT}}) \end{aligned} \tag{2}$$

where KL is the K-L scatter, IoU denotes the intersection-over-union, and MAE is the mean absolute error, calculated as shown in (3). $[p_s, b_s]$ and $[p_t, b_t]$ are the class and location predictions of a proposal for the student and teacher model, respectively, $[p_{\text{GT}}, b_{\text{GT}}]$ is the ground truth

$$\begin{aligned} \text{KL}(p, q) &= \sum_i p_i \ln\left(\frac{p_i}{q_i}\right) \\ \text{MAE}(p, q) &= \frac{1}{n}\sum_i^n |p_i - q_i| \\ \text{IoU}(\text{box}_p, \text{box}_q) &= \frac{\text{box}_p \cap \text{box}_q}{\text{box}_p \cup \text{box}_q}. \end{aligned} \tag{3}$$

Bringing (2) and (3), the equations for x_t and x_s are as (4), where n and m represent the number of old classes and the total number of classes, respectively,

$$\begin{aligned} x_t &= \left[\sum_i^n p_s(i) \ln\left(\frac{p_s(i)}{\text{soft}(p_t)(i)}\right)\right] \cdot \frac{b_s \cap b_t}{b_s \cup b_t} \\ x_s &= \left[\sum_i^m p_s(i) \ln\left(\frac{p_s(i)}{p_{\text{GT}}(i)}\right)\right] \cdot \frac{b_s \cap b_{\text{GT}}}{b_s \cup b_{\text{GT}}}. \end{aligned} \tag{4}$$

After obtaining x_t and x_s, the joint-quality q can be calculated by the following equation:

$$q = |x_t - x_s|. \tag{5}$$

According to the quality score q, the proposal selection strategy is designed with reference to the positive-negative-balance ideology [30] (prevents the detector from focusing only on the same quality proposal), and the process is shown in Algorithm 1.

For CLOD methods, the selection number of proposals normally used for knowledge distillation is 256, and we have taken the same setup for the experiments in this article.

C. Predicted Location Probability Representation

The prediction from the teacher model contains knowledge of the old classes, which we call soft-prediction. The output consists of classification and regression, where the classification output is itself a probability distribution so there is no need for transformation. Therefore, the focus of this section is on the transformation method of the regression output.

However, the classification part still requires some adaptation during CL. Before training, the dimension of the classifier first needs to be expanded, and due to the increase in the number of classes, the model needs to increase the dimension of the classification branch (usually a linear classifier in the fully-connected layer) according to the number of new classes, in the way shown in Fig. 6.

Algorithm 1 Proposal Selection Strategy

Require:
Scored proposals: $M = \{m_1, m_2, \ldots, m_n\}$
Scores from student model: $X_s = \{x_{s_1}, x_{s_2}, \ldots, x_{s_n}\}$
Scores from teacher model: $X_t = \{x_{t_1}, x_{t_2}, \ldots, x_{t_n}\}$
Final selection number: N

Ensure:
1: According to the eq. 5, the joint-quality scores $Q = \{q_1, q_2 v q_n\}$;
2: Taking the N proposals with the highest q and the N proposals with the lowest q, the sets are denoted m_h and m_l respectively;
3: Randomly select ${}^{N}/_{2}$ proposals in m_h and m_l respectively, obtain $M_h = \{m_{h1}, m_{h2}, \ldots, m_{h^{N}/_{2}}\}$ and $M_l = \{m_{l1}, m_{l2}, \ldots, m_{l^{N}/_{2}}\}$;
4: $M_N = M_h \cup M_l$
5: **return** M_N;

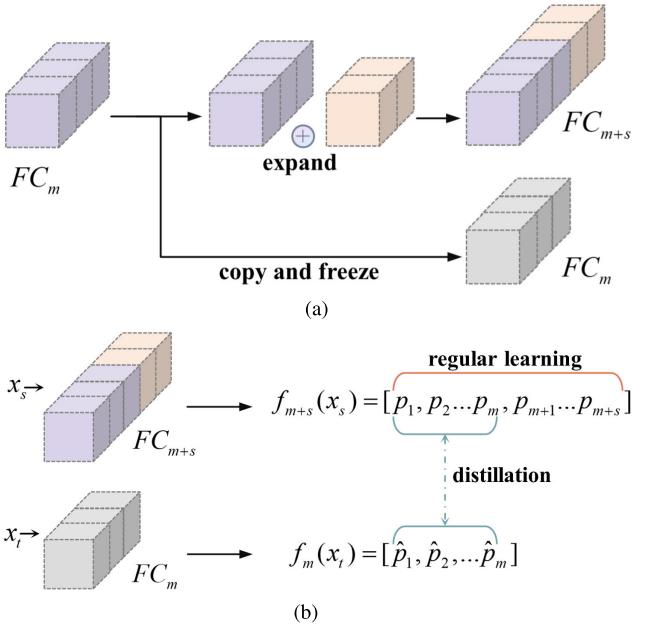


Fig. 6. Schematic of branch expansion and learning process. (a) Branch expansion when adding s classes to m classes. (b) Way of classifier learning during the CL process.

As shown in Fig. 6(b), noting the class predictions of the teacher model and the student model as $p_t = [\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_n]$ and $p_s = [p_1, p_2, \ldots, p_n, p_{n+1}, \ldots, p_{n+s}]$, respectively, n is the number of original classes, s is the number of newly added classes, and p_i is the prediction for the ith class. We soften p_t and $p_s[1 \to n]$ using the SoftMax function to obtain $S(p_t)$ and $S(p_s)$. The formula for SoftMax is in the following equation, where τ is the temperature of the hyperparameter:

$$\mathcal{S}(p) = -\frac{\exp(p/\tau)}{\sum_i \exp(p_i/\tau)}. \tag{6}$$

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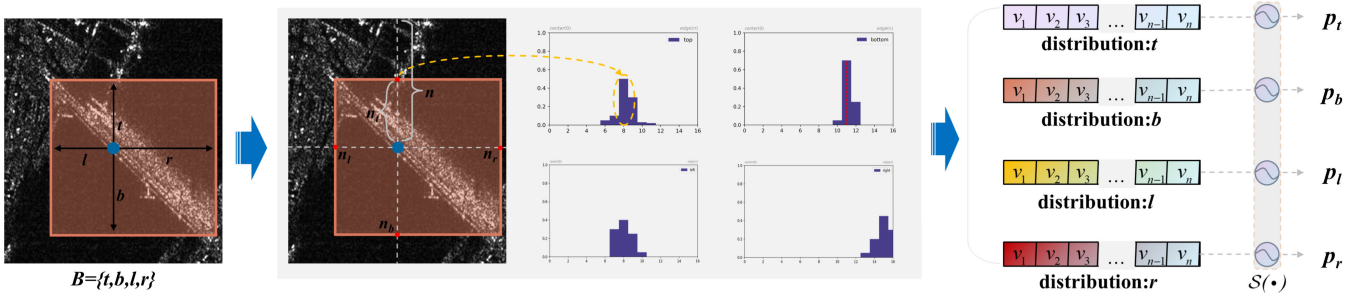


Fig. 7. Process of PLPR. First, the horizontal and vertical coordinates of the SAR image are divided into k intervals, then finding the expectation of the bounding box falling on each interval, and finally using the SoftMax function to transform the probability of the whole interval into a probability distribution with sum equal to 1.

Using the cross-entropy function for the probability distribution after SoftMax transformation, the loss of each proposal is obtained as

$$\mathrm{CE}(\mathcal{S}(p_t), \mathcal{S}(p_s)) = \sum_{i}^{n} \mathcal{S}(p_t)_i \log\left[\mathcal{S}(p_s)_i\right]. \tag{7}$$

Then, the knowledge review loss of the classified branch is

$$\mathrm{Loss}_{\mathrm{dis_cls}} = \frac{1}{N} \sum_{m} \mathrm{CE}_i \tag{8}$$

where N is the number of proposals in proposal-set m and CE_i is the cross-entropy loss of the ith proposal.

In previous work, little change was made to the localization function. There are two reasons for this: first, localization is a class-unrelated function with little performance during CL [20], [24]. Even if an unknown target is detected, the detector is equally capable of completing localization, only with reduced accuracy. Second, unlike classification, the regression output is continuous information with a distance metric, which poses a challenge for transferring regression knowledge from the teacher model to the student model.

Inspired by the work [23], this article introduces its general distribution representation method into the CL process, which in turn transforms the regression output into a probability distribution. For numerous detectors represented by fully convolutional one-stage object detection (FCOS), the predicted location of the target is denoted using $B = [t, b, l, r]$, where each of the four elements represents the distance of the sampling point from the four bounding boxes at the top, bottom, left, and right. Then, for anyone of the elements $y \in B$ has

$$y^* = \int_{d_0}^{d_{\max}} x \cdot \mathrm{Pred}(x) dx \tag{9}$$

where x denotes the regression coordinate in the interval $[d_0, d_{\max}]$ and $\mathrm{pred}(x)$ denotes the probability corresponding to that coordinate. The detection network uses the probabilities at different coordinates to describe the probability distribution of each border, and the mathematical expectation to represent the final position. In order to transform the representation of (9) into a discrete probability distribution, first the space is divided in the process shown in Fig. 7. Divide the interval $[d_0, d_{\max}]$ into k equal parts with a stride length of s, i.e.,

$$\begin{aligned} y &= [y_0, y_1, \ldots, y_k] \\ \Delta_i &= y_i - y_{i-1}, \quad |\Delta_i| = |\Delta_j| \quad \forall i, j \in \{1, 2, 3, \ldots, k\} \end{aligned} \tag{10}$$

where y denotes the divided interval with $k+1$ nodes and Δ denotes the interval between two nodes, either Δ is equal. Experiments in the original work showed the best results with $k = 16$, and this article will also follow this setup. Based on the above step, further, (9) will be transformed into a discrete form as $y_i^* = \sum_{i=1}^{k} \mathrm{Pred}(i) \cdot i$, i represents the interval in the position of y_i^*. However, since the detector will output a prediction value separately for each interval, the sum of the probabilities over the whole interval is not strictly equal to 1. Therefore, normalization is also required, and this process is implemented by the SoftMax function, which results in the final expression for the probability distribution

$$\begin{aligned} y_i^* &= \sum_{i=0}^{k} \mathcal{S}(\mathrm{pred}(i)) \cdot i \\ &\sum_{i=0}^{k} \mathcal{S}(\mathrm{pred}(i)) \cdot \Delta = 1. \end{aligned} \tag{11}$$

Following these steps, we can transform the four regression values of the location labels into four standard discrete distributions, as shown in the latter part of Fig. 7. Let $B_t = [p_t, p_b, p_l, p_r]$ and $B_s = [q_t, q_b, q_l, q_r]$ denote the probability distributions of the bounding box output from the teacher detector and the student detector, respectively, then the KL-scatter loss of the bounding box is

$$\mathrm{KL}(B_t, B_s) = \sum_{\{t,b,l,r\}} p_i \log p_i - p_i \log q_i. \tag{12}$$

Then, the knowledge review loss of the whole regression branch is

$$\mathrm{Loss}_{\mathrm{dis_reg}} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{KL}_i \tag{13}$$

where N is the number of proposals in the proposal set during each training session. By adding $\mathrm{Loss}_{\mathrm{model}}$, $\mathrm{Loss}_{\mathrm{dis_cls}}$, and $\mathrm{Loss}_{\mathrm{dis_reg}}$, the overall loss is obtained as shown in (1).

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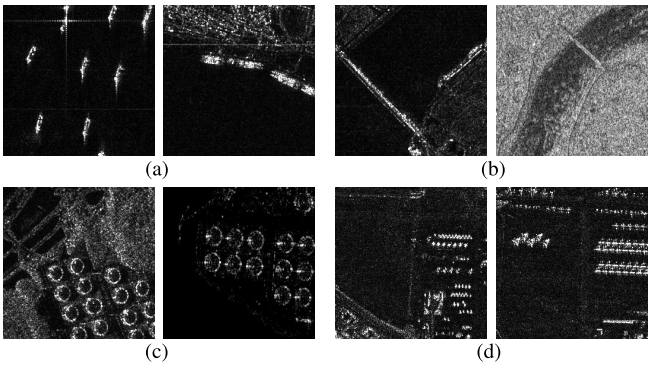


Fig. 8. Examples of the MSAR dataset, with classes (a)–(d) corresponding to ship, bridge, oilcan, and airplane, respectively.

IV. Experiment

In this section, we implement our method over the FCOS [45] network and validate our method using the MSAR-1.0 [10] dataset. First, we introduce the dataset setup, evaluation metrics, and different modes of class addition. Then, the effectiveness of each crucial component of the proposed method is evaluated in ablation experiments. Finally, the representative CLOD algorithms Faster-ILOD [20], SID [24], and ERD [22] are used as the reference for comparison experiments.

A. Implementations

1) MSAR-1.0 Dataset: In order to validate the performance of the proposed method under class continuously incremental situations, experiments are implemented on a public dataset named MSAR-1.0 [10]. The MSAR is a multiclass SAR target detection dataset released in 2023 and is currently being used in several studies. The dataset contains four classes of targets as shown in Fig. 8, respectively: ship, bridge, oilcan, and airplane, and is rich in scenarios, therefore being able to meet the requirements of this experiment while providing a challenge to detection-related tasks.

The entire dataset consists of 28 449 images and 60 456 targets, annotated using the horizontal bounding box. The data comes from HMS-1 and GF-3 satellites with polarization modes including HH, HV, VH, and VV. The details of the MSAR dataset are shown in Table I, which is an unbalanced dataset as can be seen from the number of images and targets in each class. The unbalanced data will affect the detection performance of the detector, but this is not the research of this article. Meanwhile, at each data-incremental step of CLOD experiments, only the training data for the new classes were available, but some of the images in the MSAR-1.0 dataset contained multiple class targets. Therefore, the MSAR-1.0 data was processed as follows before the experiment.

1) Images containing multiple categories in the dataset were eliminated.
2) Reduce the number of ship images to achieve a relative balance between the number of images and targets.

After processing as above, the number of images and targets in the dataset is shown in Table II.

2) Metrics: We use mAP under the cross-merge ratio of 0.5 as our main evaluation metric. AP is the area under the precision-recall curve and is calculated as in (14). mAP is the average under multiple categories of AP, and mAP = AP when the number of class is 1

$$\mathrm{AP} = \int_0^1 p(r)dr \tag{14}$$

where p represents the precision, r represents the recall, and AP is equivalent to the area under the curve. Since the detection task needs to judge the accuracy of localization by the IoU function, we usually use AP_ε to represent the AP under IoU $> \varepsilon$. In this article, we use AP_{50} by default.

3) Experimental Platform: In this article, the method is built on an FCOS detector, using Resnet-50 as the backbone and adding FPN to the neck. the whole experiment was performed on Pytorch18.2, Cuda11.3, and Cudnn8.04, using an intel core 15-9400F CPU and an Nvidia GeForce RTX 3060 graphics card.

TABLE I
Information About the Data

Sensor	HMS-1, GF-3
Wave Band	c
Polarization Mode	HH, HV, VH, VV
Resolution	5m, 3m
Image Size (pixels)	256×256, 2048×2048
Included Classes	ship, bridge, oilcan, airplane
Number of Images	ship: 26094 bridge: 1582 oilcan: 1248 airplane: 108
Number of Targets	ship: 39858 bridge: 1815 oilcan: 12319 airplane: 6368

TABLE II
Processed Dataset

	Number of Images	Number of Targets
Bridge	1253	1390
Ship	1500	2290
Airplane	101	5690
Oilcan	500	5022

B. Different Training Strategies of CL

We have multiple ways to update the old model when new data arrives. As shown in Fig. 9, for CL, we have four common training strategies.

1) Full-Data Training: As shown in Fig. 9(a), full-data training directly trains the whole model using old and new data, and there is no forgetting problem in this way. Although the training method of full-data training does not meet the conditional restriction of CL, this experiment still treats him as the upper bound for comparison.

2) Catastrophic Forgetting: As shown in Fig. 9(b), catastrophic forgetting represents the training of the entire model using only new data.

3) Fine-Tune: fine-tune usually refers to training the shared parameters of the network and the new branch using only new data, while adjusting the model parameters of the old

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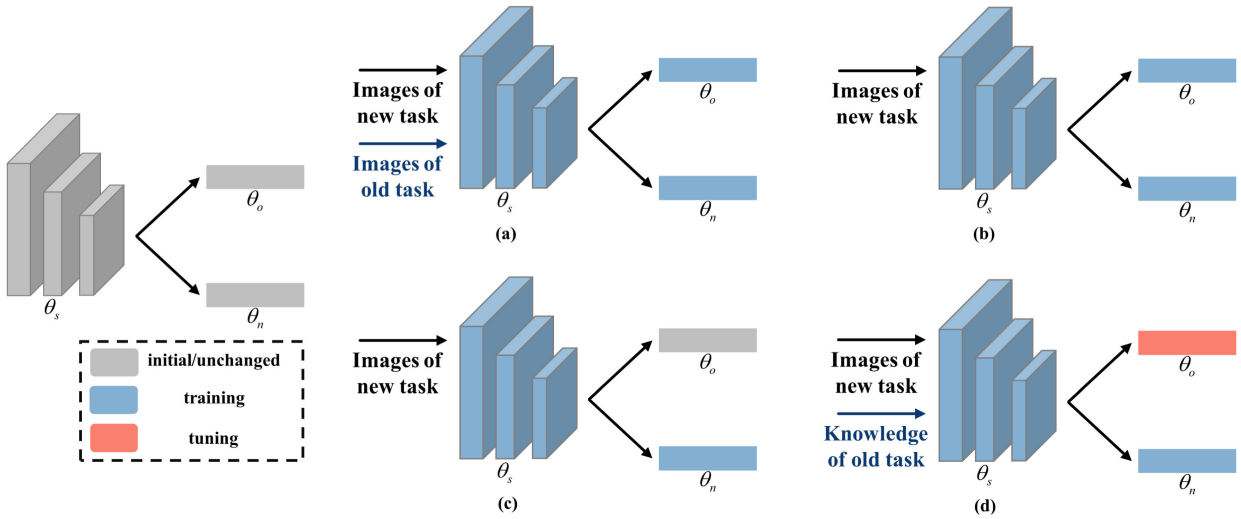


Fig. 9. Different training strategies of CL. θ_s represents the shared parameters of the network, θ_o and θ_n are the parameters of the old branch and the new branch, respectively. (a)–(d) Four different training strategies. (a) Full-data training, where the entire network is trained using all data. (b) Catastrophic forgetting, in which the entire network is trained directly using new data, and no operations are performed except for the necessary branch extensions. (c) Fine-tuning, using only new data for training while freezing the parameters of the old branches. (d) CL strategy is proposed in this article, where new data is trained while using old knowledge to maintain the capability of the old branch.

branch using fine-tune to maintain the detection capability of the network for the old data, the process is shown in Fig. 9(c). In this experiment, the model parameters of the old branch are adjusted using parameter isolation.

4) Continual Learning: CL means that the CLOD method proposed in this article trains the whole model using new classes of training samples and freezing the output of the model, ss shown in Fig. 9(d).

C. Experimental Setup

In this article, we experiment with the following CL scenarios with different settings.

1) One-Step: The model is trained using some of the classes as the base classes, and based on this, the base model is updated once using the new class data. In this experiment, we set up two one-step scenarios: i) the number of base classes is 3 and the number of new classes is 1 (subsequently shortened as 3 + 1) and ii) the number of base classes is 2 and the number of new classes is 2 (subsequently shortened as 2 + 2).

2) Multistep: The model is trained using some classes of data as the base class data, and based on this, the model is updated continuously using new classes of data step by step, with each update using new classes that have not been used. In this experiment, we use 2 classes of data as the base class data to train the model to obtain the base model, and then add one new class at a time, twice in total.

All experiments in this section were run three times with the same settings, and the final scores were averaged from the results.

D. Ablation Study

We test the effectiveness of the key components of the method proposed in this article on the MSAR-1.0 dataset. In this section, we first set different distillation proposal selection strategies based on the same detection network (FCOS). As shown in Table III, "full distillation proposal" indicates that all distillation proposals are used for distillation without selection. "Random" indicates that a portion of all the candidate proposals is selected randomly as distillation proposals, and the number of selected proposals is set to 256 in this experiment, which is the most commonly used number in CLOD and also the number of distillation proposals selected by the method proposed in this article. "Balanced" represents the balanced selection method of positive and negative proposals, in which the proposals are sorted by confidence, and 128 proposals are selected from 256 proposals with the highest confidence and 256 proposals with the lowest confidence, which together form 256 distillation proposals. "ours" represents the proposal selection strategy proposed in this article. The results from Table III show that using all distillation proposals for training is the worst, with AP of only 62.0% and 59.9% at the data-incremental settings of 2 + 2 and 3 + 1, which is consistent with the results of the analysis in this article. In contrast, the results using both the random and balanced methods are much better than the full distillation proposal method, with AP*s* of 73.4% and 68.5% for the random method and 73.7% and 68.7% for the balanced method, respectively. The distillation proposal selection method proposed in this article was the most effective among all the selection strategies, with AP of 75.9% and 69.4%. These results clearly point out the effectiveness of the

TABLE III
COMPARISON OF DETECTION RESULTS UNDER DIFFERENT DISTILLATION PROPOSAL SELECTION STRATEGIES

Method	AP_{50}(2+2)	AP_{50}(3+1)
Full-Data training	81.2	81.2
Catastrophic Forgetting	40.9	22.8
CL: Full distillation proposal	62.0	59.9
CL: Random	73.4	68.5
CL: Balanced	73.7	68.7
CL: Ours	**75.9**	**69.4**

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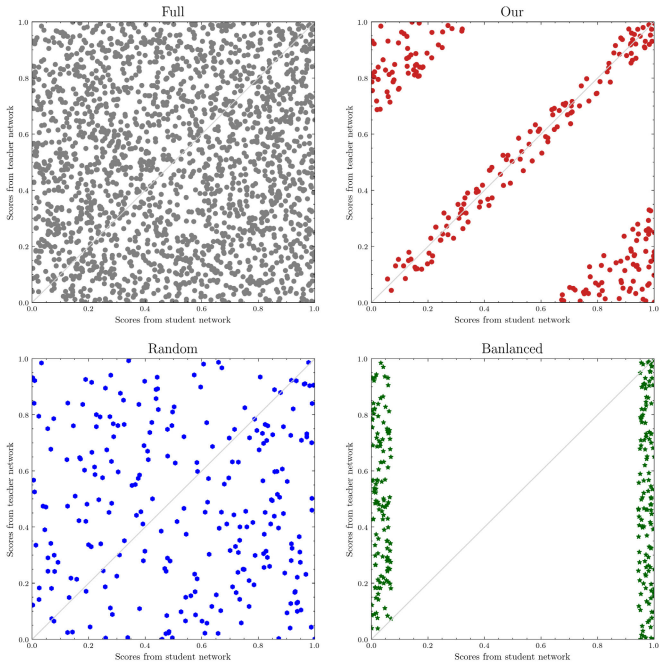


Fig. 10. Proposal distribution graph. The horizontal and vertical coordinates of each subplot are the scores of the teacher network and the student network, respectively, calculated as in (2).

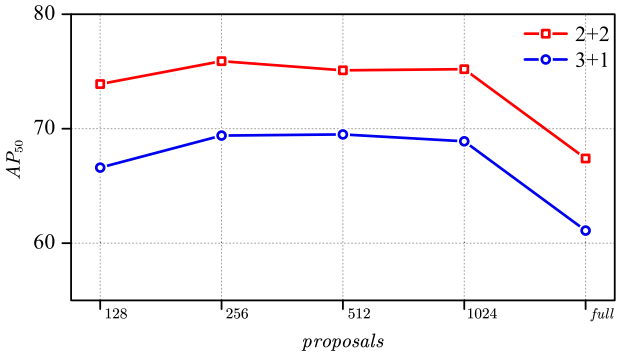


Fig. 11. Experimental results under the different number of selections of proposals.

distillation proposal selection method proposed in this article. At the same time, we draw the proposal distributions of the four CL methods. As shown in Fig. 10, where the horizontal and vertical coordinates of each scatter graph represent the scores of the teacher network and the student network.

Furthermore, we experimented with the effect of different numbers of proposals on training. We took 128, 256, 512, 1024, and full-proposal (4000) as comparison terms, using the same experimental setup. The experiments were run in both "2 + 2" and "3 + 1" scenarios using the method proposed in this article. The results are shown in Fig. 11, and it can be seen that both too few and too many proposals will make the experimental results worse. 256 and 512 settings are very close to each other, and considering the efficiency of the method, we use 256 as the number of proposals to be selected.

Compared with the previous work related to CLOD methods, this article not only distills the classification head of the detector, but also distills the regression branch (i.e., localization distillation), and the effectiveness of this section is verified. The results are shown in Table IV. "KD at cls head" means distillation is performed only on the classification branch of the detector head, while "KD at cls and reg head" is the distillation operation used in this article, and distillation is performed on both the classification branch and the regression branch. From the experimental results, it can be seen that the distillation strategy used in this article has a better performance, and the performance advantage of our method becomes more obvious as the conditions of IoU become more stringent, which is consistent with our previous analysis. A comparison of results is shown in Fig. 12.

TABLE IV
DETECTION RESULTS FOR DIFFERENT DISTILLATION STRATEGIES WITH DATA-INCREMENTAL SCENARIO SET TO 2 + 2

Method	AP_{50}	AP_{75}	AP_{80}	AP_{85}	$AP_{50:95}$
KD at cls head	75.0	46.7	28.9	9.5	45.7
Ours: KD at cls and reg head	**75.9**	**49.1**	**37.0**	**22.1**	**48.0**

In addition, we analyzed the effect of the parameter λ in the loss function on the performance. Equation (3) in λ_1 and λ_2 are used to balance the weights of classification distillation and localization distillation, respectively, and usually the weight of distillation loss in CLOD does not exceed the normal loss. We choose the values of λ_1 and λ_2 from the set (0.25, 0.5, 1) and the set (0.25, 0.5, 0.75, 1), respectively, for combination and training. Data-incremental scenes are set to 2 + 2, and the training results are shown in Fig. 13. From the results, it can be observed that our method is robust to the loss function weights, and the performance gap is not large for different λ, and on the whole, the performance decreases slightly as λ_2 increases. $\lambda_1 = 1$ and $\lambda_2 = 0.5$ is a more desirable setting, therefore, in all other experiments in this article, λ_1 and λ_2 are fixed to 1 and 0.5.

E. Comparison With Other Methods

In order to verify the advantages of the proposed method, three representative methods were chosen for comparison. The comparison methods include Faster-ILOD [20], SID [24], and ERD [22], where Faster-ILOD is based on the two-stage detector FasterRCNN [21], both SID and ERD are based on the one-stage detector FCOS [45] (the original SID was designed for CenterNet detector, and for better comparison, it is transposed to the FCOS framework in this article, denoted as SID-FCOS). We reproduced these methods and followed the settings in their papers.

1) One-Step: Table V shows the detection results under the one-step data incremental setting when the old class number is 2 and the new class number is 2. AP_{50} denotes the precision of the model for the whole data after continual training, and AP_{50}(old) and AP_{50}(new) denote the precision of the model for the old and new classes, respectively, with the difference from the upper-bound in brackets after each result value, later on we will refer to this as performance reduction. Also, we have used red boxes to circle the optimal (least performance reduction) detection results other than "full-data training" and "catastrophic forgetting." The result in Table V shows that compared to the upper-bound, the overall precision of Faster-ILOD, SID, and ERD decreased by 20.1%, 11.3%, and 6.9%, respectively, while our method performs optimally with a reduction of only 5.3%, and all CL methods are much

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TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT METHODS WHEN DATA-INCREMENTAL IS SET TO 2 + 2

Method	Basic structure	Training strategy	Base(%)	Addition(%)		
				AP_{50}	$AP_{50}(old)$	$AP_{50}(new)$
Faster-ILOD	Faster RCNN	Full-data training	79.5	80.2	79.0	81.4
		Catastrophic Forgetting		39.5 (▽50.7)	0.0 (▽79.0)	78.9 (▽2.5)
		Fine-tune		42.9 (▽37.3)	14.5 (▽64.5)	71.3 (▽10.1)
		CL		60.1 (▽20.1)	42.0 (▽37.0)	78.2 (▽3.2)
SID-FCOS	FCOS	Full-data training	81.0	81.2	80.9	81.5
		Catastrophic Forgetting		40.9 (▽40.3)	0.0 (▽80.9)	81.8 (△0.3)
		Fine-tune		37.1 (▽44.1)	1.7 (▽79.2)	72.5 (▽9.0)
		CL		69.9 (▽11.3)	72.1 (▽8.8)	67.7 (▽13.8)
ERD	FCOS	Full-data training	81.0	81.2	80.9	81.5
		Catastrophic Forgetting		40.9 (▽40.3)	0.0 (▽80.9)	81.8 (△0.3)
		Fine-tune		37.1 (▽44.1)	1.7 (▽79.2)	72.5 (▽9.0)
		CL		74.3 (▽6.9)	70.4 (▽10.5)	78.2 (▽3.3)
Proposed	FCOS	Full-data training	81.0	81.2	80.9	81.5
		Catastrophic Forgetting		40.9 (▽40.3)	0.0 (▽80.9)	81.8 (△0.3)
		Fine-tune		37.1 (▽44.1)	1.7 (▽79.2)	72.5 (▽9.0)
		CL		**75.9 (▽5.3)**	**73.7 (▽7.2)**	**78.1 (▽3.4)**

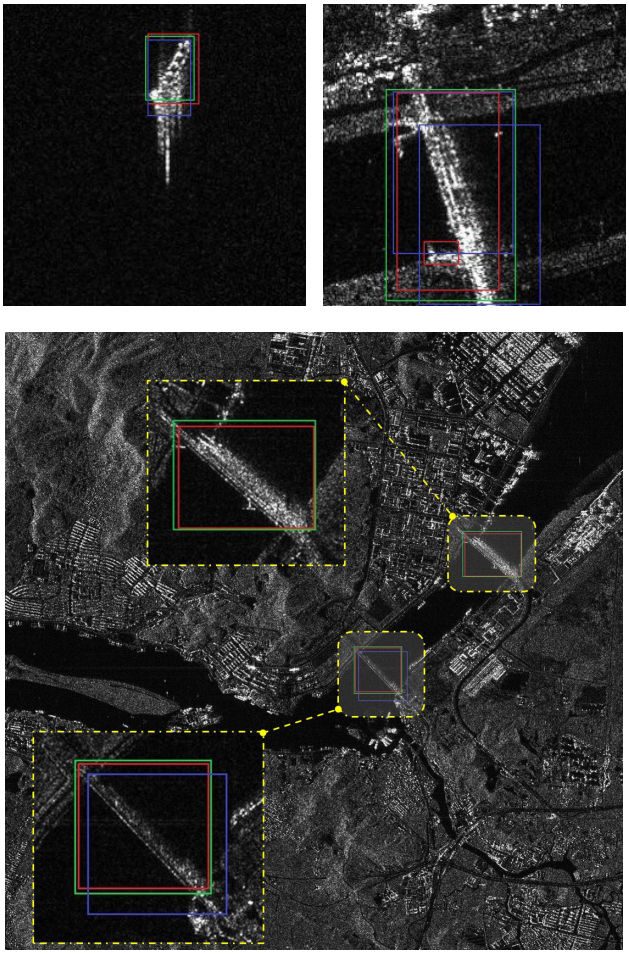


Fig. 12. Comparison of some results. The green box is the ground truth, the red box is the result of Ours, and the blue box is the result of "KD at cls head." It can be seen that the method of distillation at both cls and reg head is more accurate.

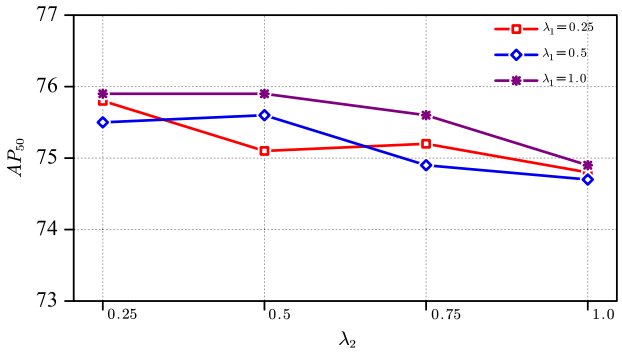


Fig. 13. Detection results for different λ values. It can be seen that our method has some robustness for λ.

better than "catastrophic forgetting" and "fine-tune," which is the same as we expected; for old classes, the performance reduction of the four methods are 37.0%, 8.8%, 10.5%, and 7.2%, respectively, and our method is still the best, which indicates that our method has ability to ease forgetfulness; in terms of new class, the best performing method is Faster-ILOD, but the difference between our method is very small, with no more than 0.2% difference between each other. Similarly, the difference with the ERD method is also quite small, thus we can assume that the three methods are almost identical. Taken together, our method has the best performance.

Meanwhile, Table VI shows the detection results under the one-step data incremental setting when the number of old classes is 3 and the number of new classes is 1. The overall results of all four methods in this scenario are worse than the "2 + 2" scenario. In this data setup, our method is still better than three other methods in terms of overall and old class precision, but not as good as Faster-ILOD in new class detection, which performs almost close to full-data training on new classes. However, in terms of overall performance, our method is still the optimal one.

2) Multistep: Table VII shows the detection results with a base class number of 2 and a new class added in two steps one by one setting. From the detection results, at the first time of increment, the ERD method performs best overall, with an AP_{50} and a AP_{50}(new) of 71.2% and 77.6%, better than our

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TABLE VI
EXPERIMENTAL RESULTS OF DIFFERENT METHODS WHEN DATA-INCREMENTAL IS SET TO $3+1$

Method	Basic structure	Training strategy	Base(%)	Addition(%)		
				AP_{50}	$AP_{50}(old)$	$AP_{50}(new)$
Faster-ILOD	Faster RCNN	Full-data training	80.5	80.2	77.1	89.6
		Catastrophic Forgetting		22.1 (▽58.1)	0.0 (▽77.1)	88.5 (▽1.1)
		Fine-tune		27.8 (▽52.4)	8.2 (▽68.9)	86.6 (▽3.0)
		CL		48.8 (▽31.4)	35.3 (▽41.8)	89.3 (▽0.3)
SID-FCOS	FCOS	Full-data training	79.9	81.2	77.8	91.4
		Catastrophic Forgetting		22.8 (▽58.4)	0.1 (▽77.7)	90.9 (▽0.5)
		Fine-tune		28.1 (▽53.1)	10.3 (▽67.5)	81.5 (▽9.9)
		CL		58.6 (▽22.6)	61.1 (▽16.7)	51.0 (▽40.4)
ERD	FCOS	Full-data training	79.9	81.2	77.8	91.4
		Catastrophic Forgetting		22.8 (▽58.4)	0.1 (▽77.7)	90.9 (▽0.5)
		Fine-tune		28.1 (▽53.1)	10.3 (▽67.5)	81.5 (▽9.9)
		CL		67.6 (▽13.6)	60.7 (▽17.1)	88.2 (▽2.7)
Proposed	FCOS	Full-data training	79.9	81.2	77.8	91.4
		Catastrophic Forgetting		22.8 (▽58.4)	0.1 (▽77.7)	90.9 (▽0.5)
		Fine-tune		28.1 (▽53.1)	10.3 (▽67.5)	81.5 (▽9.9)
		CL		**69.4 (▽11.8)**	**63.4 (▽14.4)**	**87.5 (▽3.9)**

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS WHEN DATA-INCREMENTAL IS SET TO MUTISTEP

Method	Training strategy	Base	$Step1(\%):2+1$			$Step2(\%):(2+1)+1$		
			AP_{50}	$AP_{50}(old)$	$AP_{50}(new)$	AP_{50}	$AP_{50}(old)$	$AP_{50}(new)$
Faster-ILOD	Full-data training	79.5	80.5	79.2	83.1	80.2	77.1	89.6
	Catastrophic Forgetting		28.1 (▽52.4)	0.0 (▽79.2)	84.2	22.1 (▽58.1)	0.0 (▽77.1)	88.4 (▽1.2)
	Fine-tune		36.3 (▽44.2)	15.0 (▽64.2)	78.8 (▽4.3)	25.6 (▽54.6)	5.5 (▽71.6)	86.0 (▽3.6)
	CL		56.0 (▽24.5)	44.8 (▽34.4)	78.4 (▽4.7)	36.6 (▽43.6)	19.2 (▽57.9)	88.9 (▽0.7)
SID-FCOS	Full-data training	81.0	79.9	80.5	78.7	81.2	77.8	91.4
	Catastrophic Forgetting		26.2 (▽53.7)	0.0 (▽80.5)	78.6 (▽0.1)	22.8 (▽58.4)	0.1 (▽77.7)	90.9 (▽0.5)
	Fine-tune		32.2 (▽47.7)	9.6 (▽70.9)	77.4 (▽1.3)	26.5 (▽54.7)	3.5 (▽74.3)	72.5 (▽18.9)
	CL		68.0 (▽11.9)	68.5 (▽12.0)	67.0 (▽11.7)	57.4 (▽23.8)	54.4 (▽23.4)	66.5 (▽24.9)
ERD	Full-data training	81.0	79.9	80.5	78.7	81.2	77.8	91.4
	Catastrophic Forgetting		26.2 (▽53.7)	0.0 (▽80.5)	78.6 (▽0.1)	22.8 (▽58.4)	0.1 (▽77.7)	90.9 (▽0.5)
	Fine-tune		32.2 (▽47.7)	9.6 (▽70.9)	77.4 (▽1.3)	26.5 (▽54.7)	3.5 (▽74.3)	72.5 (▽18.9)
	CL		71.2 (▽8.7)	68.0 (▽12.5)	77.6 (▽1.1)	65.9 (▽15.3)	55.6 (▽22.2)	86.6 (▽4.8)
Proposed	Full-data training	81.0	79.9	80.5	78.7	81.2	77.8	91.4
	Catastrophic Forgetting		26.2 (▽53.7)	0.0 (▽80.5)	78.6 (▽0.1)	22.8 (▽58.4)	0.1 (▽77.7)	90.9 (▽0.5)
	Fine-tune		32.2 (▽47.7)	9.6 (▽70.9)	77.4 (▽1.3)	26.5 (▽54.7)	3.5 (▽74.3)	72.5 (▽18.9)
	CL		**70.9 (▽9.0)**	**68.5 (▽12.0)**	**75.7 (▽3.0)**	**67.8 (▽13.4)**	**58.3 (▽19.5)**	**86.9 (▽4.5)**

method's 70.9% and 75.7%, but our method performs best on the old classes, with a 68.5% old class precision, better than the 68.0% of ERD, and both methods are much better than Faster-ILOD and SID. At the second time of increment, our method had a clear advantage, performing best on both overall and old classes, especially on the old class, which was far better than the other methods. In the new class, however, the best performer was still Faster-ILOD, although its performance in the old class was so poor that it was barely able to accomplish the task of resistance to catastrophic forgetting.

We show some of the experimental results, as shown in Fig. 14. The scenario is $2+2$, with the old class as ship and bridge (first four columns) and the new class as oilcan and airplane (last four columns), each row represents the results of different training methods. It can be seen that our method performs best for detecting the old classes, but still has some false alarms and missed detection compared to the full-data training.

Combining the results of one-step and multistep shows the effectiveness of the method proposed in this article, which achieves the best results in a variety of CL scenarios. However, the performance is not optimal in some cases, which we will discuss in more detail in the next section.

V. DISCUSSION

In the experiments conducted in the preceding section, we initially validated the efficacy of the fundamental components of the proposed method. Subsequently, we established several CLOD scenarios and illustrated the method's capability to mitigate forgetting through comparative analysis in each

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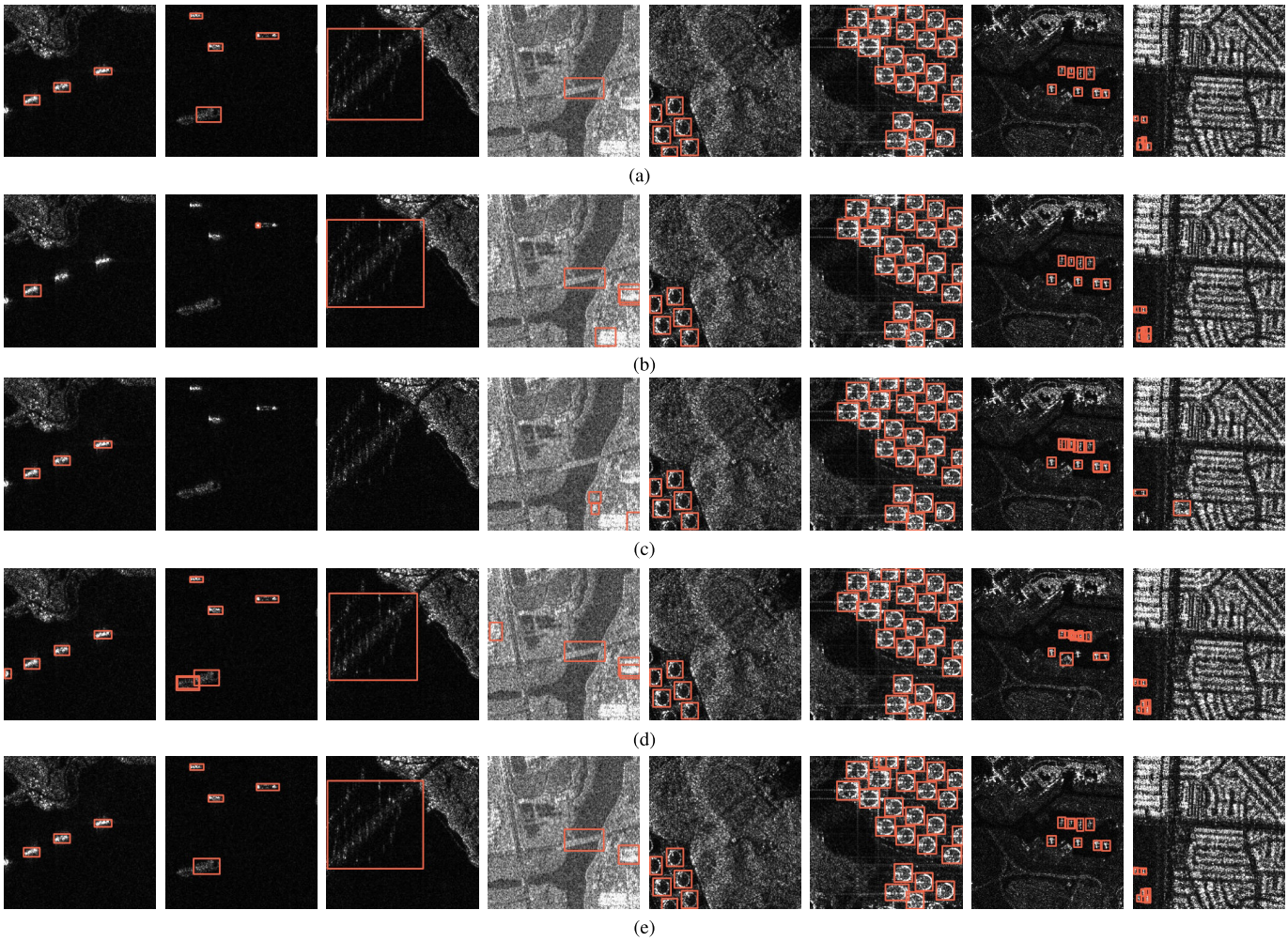


Fig. 14. Comparison of experimental results. In this case, the first four columns are the old class of ship and bridge, and the last four columns are the new class of oilcan and airplane. Each row means a different training strategy, for (a) full-data training, (b) CLOD method using Faster-ILOD, (c) SID-FCOS, (d) ERD, and (e) method proposed in this article. Except for fasterILOD which uses fasterRCNN as the basic structure, all the others are based on FCOS.

scenario. Nevertheless, our experimental findings uncovered certain issues warranting discussion.

First, in the comparison experiments, we observed that all methods exhibited inferior performance in the "3 + 1" scenario compared to the "2 + 2" scenario. This implies that the scenario with an equal number of old and new classes outperformed the scenario with fewer new classes than old classes. We speculate that this phenomenon is akin to the issue of "sample imbalance" encountered in traditional training settings. While previous research in object classification has highlighted the adverse impact of unbalanced sample sizes on learning [46], [47], these studies have predominantly focused on conventional training and approached the issue from a numerical rather than a class-oriented perspective. In the area of object detection, the problem of sample imbalance or small samples still exists [48], [49], and the result is that classes with small samples are usually poorly detected, so these works focus on improving the detection of small sample classes. For conventional training, imbalance can be defined in terms of objective sample numbers, however, for distillation, is there also an imbalance between knowledge (model parameters) and data From the experimental results, this imbalance does exist, so how to define the imbalance between knowledge and data and how to solve the negative effects of this imbalance is the next question we need to consider.

Second, our method exhibits certain limitations. Across all comparison experiments, our method demonstrates slightly lower learning efficacy in new classes compared to other methods. This suggests a negative impact of the proposed method on the learning of new classes by the SAR target detector, despite its ability to enhance memory. Addressing this issue constitutes the next step in our research agenda.

Finally, we have given some thought to the practicality of the method. The process of distillation necessarily requires some time consumption as well, so overall the advantages of a distillation-based method for CLOD are not obvious at a small data scale. We made a preliminary observation on multistep incremental trials, recording the time consumption of the full-data training strategy and our method, and the results are shown in Fig. 15. The results are in line with our expectations, at the beginning of the data incremental phase CLOD method does not show the efficiency advantage, but as the data continuously increments, the efficiency of the CLOD method

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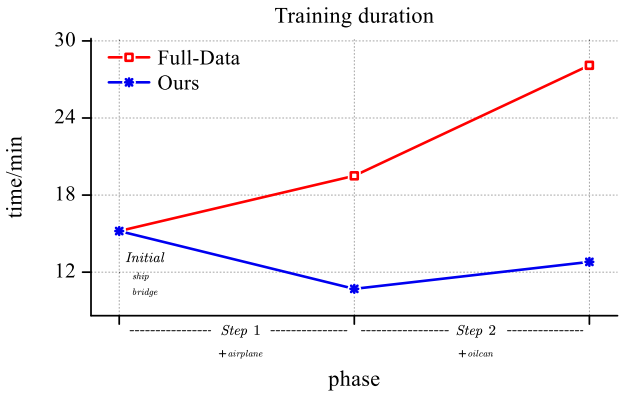


Fig. 15. Time consumption of different training strategies. The data scenario is "2 + 1 + 1" multistep data increment, each training is fixed for 12 epochs, and the average time consumption per epoch is recorded for each training phase. The red and blue lines represent the full data training and our method, respectively, and it can be seen that with the gradually increasing data, the efficiency advantage of the CLOD method becomes more and more significant.

becomes more and more obvious. It is foreseeable that as the data scale gradually increases, the time consumption of the CLOD method will be far lower than that of full data training. There is a more realistic enhancement scheme for KD-based CLOD methods, as the old model parameters are frozen and the feature extraction for the images is fixed, so the feature extraction results for all images can be retained in the first loop of the training process and called directly in subsequent loops. However, we have only conducted a methodological study in this article, and therefore no implementation of this solution has been carried out.

VI. Conclusion

This article delves into the phenomenon of catastrophic forgetting in SAR target detection and explores the similarities and differences between SAR and optical images concerning classification and localization. In addition, we identify the inadequacies of previous CL methods for SAR target incremental detection, particularly in addressing the challenges posed by significant background variations in SAR images. As a result, we propose a focused CL method tailored specifically for SAR target detection. This method is built upon distillation and introduces a PLPR technique, which converts regression outputs into probability distributions during knowledge review, thus mitigating the degradation of localization performance caused by diverse scenarios. Furthermore, we observe that the selection of proposals during the knowledge transfer process significantly influences CL performance. To further enhance the effectiveness, we design a proposal selection strategy tailored to the characteristics of SAR images.

We set up multiple kinds of data increment scenarios on the MSAR-1.0 dataset, and the method performs best in most scenarios, meeting the purpose of relieving catastrophic forgetting according to the experimental results. However, it should also be noted that although our method has good performance, the results in multistep addition scenarios are still a gap to full-data training, which needs further analysis and research.

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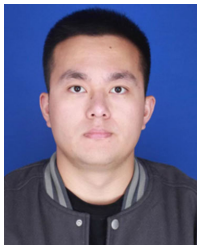
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